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Gestão autônoma de redes baseada em intensão (*intent-based networking*) usando LLMs e arquiteturas multi-agente

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Natal (RN), maio de 2026

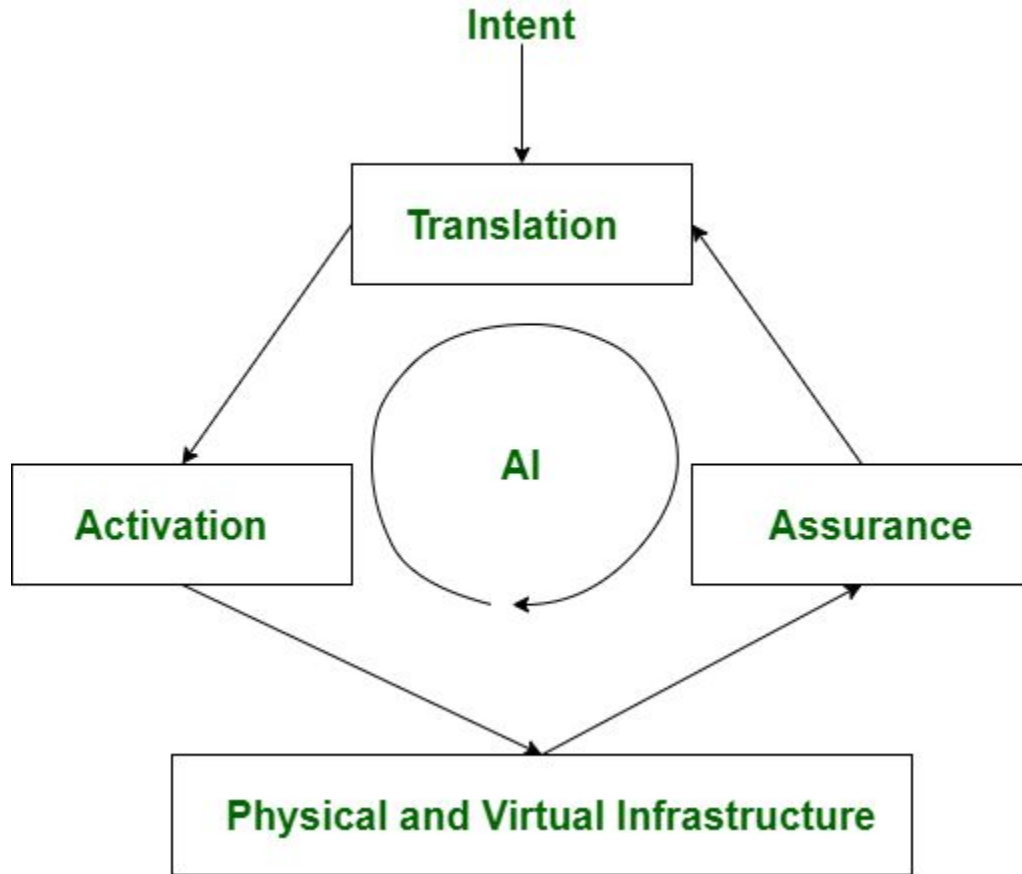


Problema

- O que está sendo resolvido?

Intent-Based Networking (IBN) é um método de **simplificação da gestão de redes** por meio de análise e aplicação da intenção do usuário, abstraindo trabalho de configuração e ação manual específica.

Exemplo prático: haverá um evento de *e-sports* no(s) laboratório(s) da universidade, ao invés de precisar revisar e ajustar o fluxo de dados (*bps*) para cada porta de cada dispositivo (*switch/firewall*), o gestor pode solicitar ao agente que “prepare os laboratórios X e Y para o evento”.



Problema

- **Por que isso é relevante?**

Em especial (mas não limitado a):

- a. **Automação** de tarefas massantes e repetitivas, como busca, interpretação e configuração de dispositivos (especialmente em redes grandes)
- b. **Redução de erro humano**, ao automatizar mudanças granulares de regras
- c. **Alocação mais eficiente de recursos**. Em sistemas multi-agente, estes podem negociar recursos afim de reduzir custos e maximizar qualidade de serviço (QoS)

Abordagem

- Como as ferramentas foram utilizadas?

As ferramentas foram usadas para extrair *insights* cruzados entre os seguintes arquivos, incluindo **artigos científicos**, **documentação técnica** e **posts de blog especializados**:

Name	Size
 2021-07-13 RIC-xApp-rApp players.pdf	214,7 kB
 2403.02238v2.pdf	624,9 kB
 2406.06059v2.pdf	2,3 MB
 2505.01841v1.pdf	4,8 MB
 2508.06616v2.pdf	1,3 MB
 2601.13114v1.pdf	4,2 MB
 985646264-Towards-Intent-based-Network-Management-for-the-6G-System-Ado...	595,7 kB
 Maestro_LLM-Driven_Collaborative_Automation_of_Intent-Based_6G_Networks....	2,2 MB
 Mobile-LLaMA.pdf	1,4 MB
 s11432-024-4337-1.pdf	9,3 MB
 s13638-025-02472-x.pdf	1,8 MB
 ts_123288v160400p.pdf	562,9 kB

Abordagem

- Como isso se conecta ao tema escolhido?



RIC, xApps, rApps: Who are the key players?

As the radio access network continues to evolve towards more open architecture, opportunities for innovation in the Open RAN space have proliferated. One of the central components within Open RAN architecture is the RAN Intelligent Controller (RIC), which provides an open hosting platform for innovative xApps and rApps. We highlight 12 key players in this space.

Grace Donnelly, Senior Consultant



ETSI TS 123 288 V16.4.0 (2020-07)



5G;
Architecture enhancements for 5G System (5GS)
to support network data analytics services
(3GPP TS 23.288 version 16.4.0 Release 16)



INTERNATIONAL CONFERENCE ON THE DESIGN OF RELIABLE COMMUNICATION NETWORKS

Towards Intent-Based Network Management: Large Language Models for Intent Extraction in 5G Core Networks

Research-driven network automation through sustainable multimodal generative AI

IntAgent: NWDAF-Based Intent LLM Agent Towards Advanced Next Generation Networks

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Abstract—Intent-based networks (IBNs) are gaining prominence as an innovative technology that automates network operations through high-level request statements, defining what the network should achieve. In this work, we introduce IntAgent, an intelligent intent LLM agent that integrates NWDAF analytics and tools to fulfill the network operator's intents. Unlike previous approaches, we develop an intent tools engine directly within the NWDAF analytics engine, allowing our agent to utilize live network analytics to inform its reasoning and tool selection. We offer an enriched, 3GPP-compliant data source that enhances the dynamic, context-aware fulfillment of network operator goals, along with an MCP tool server for scheduling, monitoring, and analytics tools. We demonstrate the efficacy of our framework through two practical use cases: ML-based traffic prediction and scheduled policy enforcement, which validate IntAgent's ability to autonomously fulfill complex network intents.

Index Terms—Core Network, Next Generation Network, Large Language Models, Intent Management, Intelligent Networks, Closed Loop.

I. INTRODUCTION

Rapid surge in network traffic and exponential growth of connected devices are driving the development of next-generation networks (NGNs), including 6G and beyond [1]. 5G has already transformed network require-

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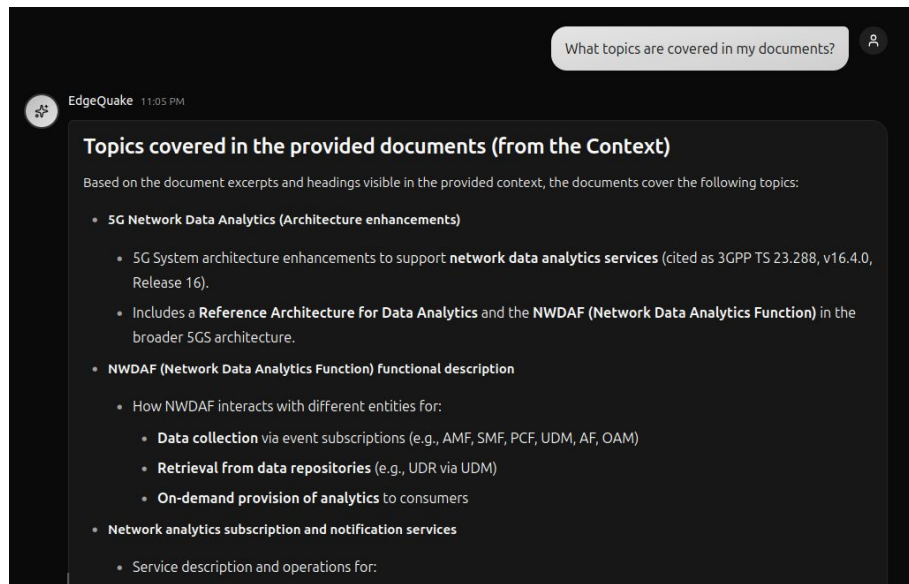
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[†] Qatar University, Doha, Qatar.

EdgeQuake

- Configurações
 - gpt-5-mini
 - text-embedding-3-small
- Processados 12 documentos
 - Artigos, documentação técnica, posts de blogs especializados
 - 10718 entidades; 11602 relacionamentos; 13 categorias
- 477 requisições e ~1.9M tokens
- Custo total (real): \$2.04
- Tempo total de processamento: ~12min

Evidências

- Resultados obtidos (*outputs*, exemplos, demonstrações)



What topics are covered in my documents?

EdgeQuake 11:05 PM

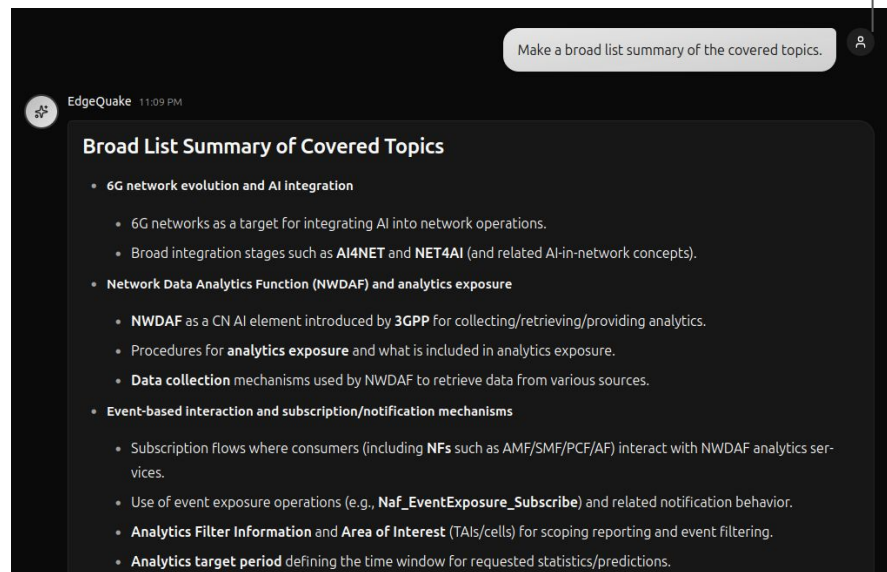
Topics covered in the provided documents (from the Context)

Based on the document excerpts and headings visible in the provided context, the documents cover the following topics:

- **5G Network Data Analytics (Architecture enhancements)**
 - 5G System architecture enhancements to support **network data analytics services** (cited as 3GPP TS 23.288, v16.4.0, Release 16).
 - Includes a **Reference Architecture for Data Analytics** and the **NWDAF (Network Data Analytics Function)** in the broader 5GS architecture.
- **NWDAF (Network Data Analytics Function) functional description**
 - How NWDAF interacts with different entities for:
 - **Data collection** via event subscriptions (e.g., AMF, SMF, PCF, UDM, AF, OAM)
 - **Retrieval from data repositories** (e.g., UDR via UDM)
 - **On-demand provision of analytics** to consumers
- **Network analytics subscription and notification services**
 - Service description and operations for:

Observa-se conexão entre os tópicos 5G e novos serviços de análise de dados, além do NWDAF

Observa-se sobreposição entre os tópicos 6G e integração de IA preditiva em redes (AI4NET)



Make a broad list summary of the covered topics.

EdgeQuake 11:09 PM

Broad List Summary of Covered Topics

- **6G network evolution and AI Integration**
 - 6G networks as a target for integrating AI into network operations.
 - Broad integration stages such as **AI4NET** and **NET4AI** (and related AI-in-network concepts).
- **Network Data Analytics Function (NWDAF) and analytics exposure**
 - **NWDAF** as a CN AI element introduced by **3GPP** for collecting/retrieving/providing analytics.
 - Procedures for **analytics exposure** and what is included in analytics exposure.
 - **Data collection** mechanisms used by NWDAF to retrieve data from various sources.
- **Event-based interaction and subscription/notification mechanisms**
 - Subscription flows where consumers (including **NFs** such as AMF/SMF/PCF/AF) interact with NWDAF analytics services.
 - Use of event exposure operations (e.g., **Naf_EventExposure_Subscribe**) and related notification behavior.
 - **Analytics Filter Information** and **Area of Interest** (TAIs/cells) for scoping reporting and event filtering.
 - **Analytics target period** defining the time window for requested statistics/predictions.

Evidências

“What's the importance of using agents/LLM for network management?”

O modelo evidencia novas conexões com IBN, agentes e raciocínio em tempo real, usando ferramental NWDAF.

Destaca relevância de *frameworks* agente multi-agente existentes, como o IntAgent e o MAESTRO.

3) Provide trustworthy, data-driven reasoning via analytics (e.g., NWDAF integration)

- **NWDAF (Network Data Analytics Function)** is described as a standardized way to integrate AI into the core network. It acts as a **centralized analytics engine**, collecting data from network functions, user equipment, and management systems to support intelligent automation.
- **IntAgent** is highlighted as an “intelligent intent LLM agent” that **integrates NWDAF analytics and tools**, where the agent uses **live network analytics** to inform reasoning and tool selection.
- The context also emphasizes that previous approaches may lack **deep integration with network analytics functions such as NWDAF** for real-time, data-driven decisions.

4) Support autonomous and closed-loop actions using tool-using agents

- In the IntAgent description, the agent is built to use an **intent tools engine** within the NWDAF analytics engine, and it uses tools to fulfill intents (with examples like **ML-based traffic prediction** and **scheduled policy enforcement**).
- The context further frames LLMs as evolving into an **agentic function**: they can plan and execute actions on their own to achieve a desired goal—important for IBN where the system must translate intent into configurations/actions.

5) Improve coordination in multi-stakeholder settings (multi-agent negotiation/mediation)

- **MAESTRO** is described as an LLM-driven collaborative plane/framework for automation and collaboration, where an ongoing negotiation process mediates between multiple “Business Operators” (agents).
- This matters because network management can involve **conflicting objectives** and the need for a consensus/reconciliation mechanism.

Evidências

Now focusing on using agents/LLMs in IBN. Based on what you have as a knowledge base, what are some gaps in information you think you might have? What do you think is missing? How would you start complementing your own knowledge base in this topic?

EdgeQuake 12:42 AM

What gaps in information might you have (based on the provided knowledge base)

1) Precise “tool creation” and standardized NWDAF pipeline coverage

The context notes that prior systems “**rarely address tool creation based on standardized analytics or dynamic use of live network data**” and highlights a gap in the NWDAF pipeline. However, while IntAgent is described as bridging this (with an “intent tools engine directly within the NWDAF analytics engine”), the knowledge base does **not** spell out, in detail, *how* standardized analytics are mapped to specific tool definitions across all cases (e.g., full coverage of tool creation, extensibility rules, and end-to-end pipeline guarantees).

2) Trustworthiness details beyond high-level “safety tools”

You have several components mentioned at a high level:

- assumptions blocking,
- goal tracking,
- structured validation of tool calls/parameters,
- safety gatekeeping asking for human approval,

Questionando sobre “*gaps*” na própria base de conhecimentos assimilada:

- Doc de padronização da *pipeline* NWDAF
- Metodologias de avaliação de segurança
- Detalhamento sobre colaboração multi-agente
- ... etc

Isso pode ser feito iterativamente para encontrar *gaps* literários.

Evidências

Aqui, o modelo sugere como complementar a própria base de conhecimento:

How I would start complementing your knowledge base

1. Add a "capability-to-requirements" mapping layer

- For each IBN capability (intent extraction, translation, analytics retrieval, tool execution, verification, notification), store:
 - required inputs/outputs,
 - dependencies on NWDAF operations (e.g., subscription/notify, event exposure),
 - safety checks needed before execution.

2. Extend coverage with evaluation artifacts

- Add:
 - test scenarios (more than the two described use cases),
 - quantitative metrics for correctness/safety,
 - comparison records across multiple LLM choices and prompt/system-prompt variants,
 - logging/trace formats that let you assess where hallucinations occur in the loop.

3. Model the "agentic loop" with explicit safety contracts

- Store formalized rules for the steps already described (plan → tool-call → observation → interpret/update context → finalize), including:

"Documentação de capacidades IBN (input/outputs)"

"Delimitação de artefatos de avaliação: testes, métricas, formatação para logs, etc"

"Formalização de regras para ações de agentes: planejamento, chamada de ferramenta, observação, etc"

Limitações (EdgeQuake)

Vantagens:

- Pode ser *self-hosted*, “possibilitando” uso de dados sensíveis
 - Dados processados pela OpenAI API não são (alegadamente) usados para treinamento de novos modelos*
- Respostas são mais técnicas, referenciáveis e pouco verbosas

Desvantagens:

- Abrir e fechar o programa excluía as configurações do modelo selecionado
- Falta de mais formatos para exportação de grafos (.html)

Dificuldades:

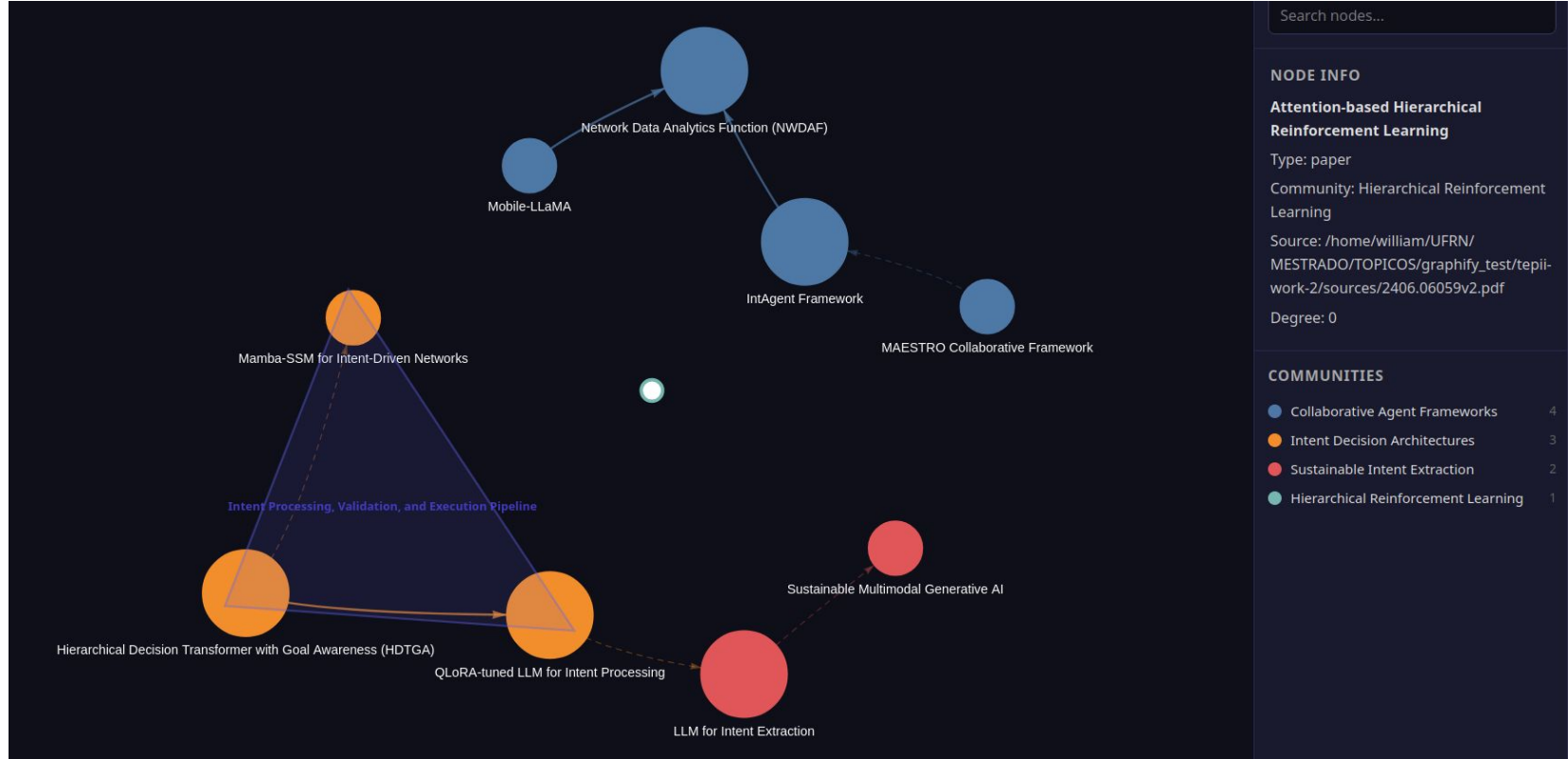
- Documentos grandes (+50 páginas) falharam e tiveram que ser reiniciados
- Aba de “custos” não reflete o custo real da API (\$0.742 vs \$2.04), nem a de “total tokens” (1.4M vs 1.9M)

* <https://developers.openai.com/api/docs/concepts>

Graphify

- Configurações (Codex)
 - gpt-5.4-mini
 - Reasoning Level: Medium
- Processados 12 documentos
 - Artigos, documentação técnica, posts de blogs especializados
 - 10 entidades; 7 relacionamentos; 4 categorias
 - Extracted: 43%
 - Infered: 57% (confiança média de 0.86)
 - Ambiguous: 0%
- 22 requisições e ~1.3M tokens
- Custo total (geração): \$0.24

Knowledge graph - Graphify



Evidências

- Resultados obtidos (outputs, exemplos, demonstrações)

```
> $graphify query "o que é MCP?"
```

- Vou pegar a definição do MCP no paper e a conexão dele com o IntAgent para responder de forma precisa, sem misturar com a definição genérica de MCP fora do corpus.

- Ran `python3 - <<'PY'`

```
| import json
| from pathlib import Path
| ... +8 lines
| {
|   "relation": "conceptually_related_to",
|   ... +17 lines (ctrl + t to view transcript)
|   "target": "ts_123288v160400p_nwdaf"
| }
```

- Ran `pdftotext '/home/william/UFRN/MESTRADO/TOPICOS/graphify_test/tepii-work-2/sources/2601.13114v1.pdf' - | rg -n`

```
| -i 'Model Context Protocol|MCP|client-server architecture|tools'
| 15:that integrates NWDaf analytics and tools to fulfill the
| 17:we develop an intent tools engine directly within the
| ... +47 lines (ctrl + t to view transcript)
782:[17] X. Hou, Y. Zhao, S. Wang, and H. Wang, "Model context protocol (mcp): Landscape, security threats, and
future research
```

*O modelo vai mostrando
o caminho de raciocínio
e busca no grafo.*

Evidências

- Resultados obtidos (outputs, exemplos, demonstrações)

• No contexto do corpus, MCP é o **Model Context Protocol**: um protocolo usado para conectar ferramentas a agentes de IA em uma arquitetura **client-server**. O paper do IntAgent diz que o MCP unifica a integração das ferramentas em vez de exigir integrações separadas para cada API ou serviço, e que o IntAgent usa isso como base para acessar ferramentas de **agendamento, monitoramento e analytics**. /home/william/UFRN/MESTRADO/TOPICOS/graphify_test/tepii-work-2/sources/2601.13114v1.pdf

No paper, o MCP aparece como a camada que permite ao agente:

- descobrir ferramentas,
- chamá-las como um cliente MCP,
- e operar com uma estrutura mais padronizada e segura.

O grafo não extraiu um nó separado para MCP, mas mostrou que ele está embutido na arquitetura do /home/william/UFRN/MESTRADO/TOPICOS/graphify_test/tepii-work-2/sources/2601.13114v1.pdf, que por sua vez **references** /home/william/UFRN/MESTRADO/TOPICOS/graphify_test/tepii-work-2/sources/ts_123288v160400p.pdf. Ou seja: no corpus, MCP não é o objetivo final, e sim o mecanismo de ligação entre o LLM e o conjunto de ferramentas operacionais do IntAgent. /home/william/UFRN/MESTRADO/TOPICOS/graphify_test/tepii-work-2/graphify-out/GRAPH_REPORT.md

Se quiser, posso **agora**:

1. explicar **como o MCP aparece no fluxo do IntAgent**,
2. ou comparar **MCP com APIs tradicionais** no contexto do paper.

Apresenta a resposta da requisição

Posiciona a resposta no grafo e no relatório

Evidências

- Resultados obtidos (GRAPH_REPORT.md)

Knowledge Gaps

- **5 isolated node(s):** Mobile-LLaMA, MAESTRO Collaborative Framework, Attention-based Hierarchical Reinforcement Learning, Mamba-SSM for Intent-Driven Networks, Sustainable Multimodal Generative AI These have ≤ 1 connection - possible missing edges or undocumented components.
- **Thin community Sustainable Intent Extraction** (2 nodes): LLM for Intent Extraction, Sustainable Multimodal Generative AI Too small to be a meaningful cluster - may be noise or needs more connections extracted.
- **Thin community Hierarchical Reinforcement Learning** (1 nodes): Attention-based Hierarchical Reinforcement Learning Too small to be a meaningful cluster - may be noise or needs more connections extracted.

Suggested Questions

Questions this graph is uniquely positioned to answer:

- **Why does QLoRA-tuned LLM for Intent Processing connect Intent Decision Architectures to Sustainable Intent Extraction?** High betweenness centrality (0.111) - this node is a cross-community bridge.
- **Why does LLM for Intent Extraction connect Sustainable Intent Extraction to Intent Decision Architectures?** High betweenness centrality (0.083) - this node is a cross-community bridge.
- **Are the 2 inferred relationships involving LLM for Intent Extraction (e.g. with QLoRA-tuned LLM for Intent Processing and Sustainable Multimodal Generative AI) actually correct?** LLM for Intent Extraction has 2 INFERRED edges - model-reasoned connections that need verification.
- **What connects Mobile-LLaMA, MAESTRO Collaborative Framework, Attention-based Hierarchical Reinforcement Learning to the rest of the system?** 5 weakly-connected nodes found - possible documentation gaps or missing edges.

Limitações (Graphify)

Vantagens:

- Transparência
- GRAPH_REPORT
- Os dados gerados são multiplataforma (codex, gemini, cloud code...)
- Se além ao grafo

Desvantagens:

- Esse não é o objetivo principal da ferramenta
- Baixa extração de Conhecimento para o Grafo (para essa base)
- Não é muito boa como ferramenta de aprendizado

Dificuldades

- Sem muitas dificuldades (para essa base)
- Limitações do grafo
- As não consegue interpretar o documento totalmente

NotebookLM

Vantagens:

- Grátis
- Carregamento mais rápido (< 1min)
- Criação de mapa mental, slide, resumo em podcast, vídeo, etc
- Opinião: *output* mais organizado, mas mais verboso

Desvantagens:

- Não permite *self-hosting*
 - Documentos ficam carregados nos servidores do Google
- Apenas o uso de modelos Gemini (Gemini 3 Flash/Pro)

Obrigado!
Dúvidas?